GO, KEGG, and GSEA Enrichment Analysis Workflow

Brian Walker
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1 Overview

What is This? This is a complete, step-by-step guide to analyze your RNA-seq data using three different enrichment methods: GO, KEGG, and GSEA. Each method provides different biological perspectives on which pathways are affected in your experiment.

1.1 Assumptions

- You have completed DESeq2 analysis
- You have saved results with gene symbols as rownames

1.2 Input Requirements

Your DE results must contain:

Column	Description
gene	Gene symbol (rownames)
log2FoldChange	Effect size
pvalue	Statistical significance
padj	FDR-adjusted p-value

Example: ow_de

2 Package Installation and Loading

What is This? We need various packages from Bioconductor and CRAN for the enrichment analysis. These packages contain functions to access gene ontology databases.

```
# Install BiocManager if needed
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

# Install packages (run once)
BiocManager::install(c("clusterProfiler",
                       "org.Hs.eg.db",
                       "enrichplot",
                       "DOSE",
                       "pathview",
                       "ggplot2",
                       "dplyr",
                       "stringr",
                       "ggridges"))

# Load packages
library(clusterProfiler)
library(org.Hs.eg.db)
library(enrichplot)
library(DOSE)
library(pathview)
library(ggplot2)
library(dplyr)
library(stringr)
library(ggridges)
```

3 Data Loading

What is This? We load the differential expression results file. This file contains information about which genes changed between your experimental groups.

3.1.1 Read the DE results

```
# Read the DE results
# Assumes you have saved results with gene symbols as rownames
# Example: write.csv(as.data.frame(ow_de), "ow_de_results.csv")
ow_de <- read.csv("ow_de_results.csv", row.names = 1)

# Check what we loaded
head(ow_de, 3)
```

```
     gene baseMean log2FoldChange     lfcSE  stat ...1 pvalue ...11 padj
ZNF683 ZNF683 1127.9       1.1124 0.1179... 9.4... NA 0.0000...  NA 0.0...
MYOM2   MYOM2   29.3       1.0127 0.2157... 4.6... NA 0.0000...  NA 0.0...
HES4     HES4  297.4       1.0101 0.1127... 8.9... NA 0.0000...  NA 0.0...
TNFRSF4 TNFRSF4   8.6       0.9917 0.3577... 2.7... NA 0.0066...  NA 0.0...
KLHDC7B KLHDC7B 1347.7       0.9911 0.1179... 8.4... NA 0.0000...  NA 0.0...
RNF207  RNF207  1108.8       0.9811 0.1179... 8.3... NA 0.0000...  NA 0.0...
```

```
dim(ow_de)
```

```
[1] 16,204     7
```

```
# See column names
colnames(ow_de)
```

```
[1] "gene"           "baseMean"       "log2FoldChange" "lfcSE"
[5] "stat"           "...1"           "pvalue"         "...11"
[9] "padj"
```

```
# Check for required columns
"gene" %in% colnames(ow_de)
"log2FoldChange" %in% colnames(ow_de)
"pvalue" %in% colnames(ow_de)
"padj" %in% colnames(ow_de)
```

```
[1] TRUE
[1] TRUE
[1] TRUE
[1] TRUE
```

4 Data Preparation

What is This? We create up-to-date versions of gene lists from Ensembl using biomaRt. This ensures our analysis uses gene relative data.

```
# Create up-to-date gene lists
ensembl_genes <- ow_de$gene
cat("Using", length(ensembl_genes), "genes\n")
```

```
Using 16,204 genes
```

```
# See first few genes
head(ensembl_genes)
```

```
[1] "ZNF683" "MYOM2"  "HES4"   "TNFRSF4" "KLHDC7B" "RNF207"
```

5 Analysis 1: GO Enrichment

What is This? Gene Ontology (GO) enrichment analysis tests biological pathways, molecular functions, or cellular components to see if more genes than expected are in your list. It tells you which biological processes are significantly affected.

GO Input Parameters Analysis

5.1 Select DEGs

```
# Create DEG list
sig_genes <- ow_de[ow_de$padj < 0.05]
cat("Found", length(sig_genes), "significant genes\n")
```

```
Found 0 significant genes
```

5.2 Run GO Enrichment

```
ego_bp <- enrichGO(
  gene = sig_genes,
  OrgDb = org.Hs.eg.db,
  keytype = "SYMBOL",
  ont = "BP",
  pAdjustMethod = "BH",
  pvalueCutoff = 0.05,
  qvalueCutoff = 0.2)

head(ego_bp@result)
```

```
                   ID                                   Description GeneRatio
GO:0006955 GO:0006955                                      immune ...        ...
GO:0002376 GO:0002376                                      immune ...        ...
GO:0006952 GO:0006952                                      defense ...        ...
GO:0050896 GO:0050896                                      response ...        ...
GO:0009605 GO:0009605                                      response ...        ...
GO:0051716 GO:0051716                                      cellular ...        ...
                  BgRatio       pvalue     p.adjust       qvalue
GO:0006955 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0002376 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0006952 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0050896 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0009605 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0051716 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
                geneID Count
GO:0006955 ZNF683/...  1347
GO:0002376 ZNF683/...  1347
GO:0006952 ZNF683/...  1347
GO:0050896 ZNF683/...  1347
GO:0009605 ZNF683/...  1347
GO:0051716 ZNF683/...  1347
```

5.3 GO Enrichment Results Table

What is This? This shows the top 10 biological processes that are most significantly changed genes. Each row is a biological process, ordered by how significant the enrichment is.

```
# Show top 10 results
head(ego_bp@result, 10)[, c("ID", "Description", "GeneRatio", "BgRatio", "pvalue", "p.adjust", "qvalue")]
```

```
                   ID                                   Description GeneRatio
GO:0006955 GO:0006955                                      immune ...        ...
GO:0002376 GO:0002376                                      immune ...        ...
GO:0006952 GO:0006952                                      defense ...        ...
GO:0050896 GO:0050896                                      response ...        ...
GO:0009605 GO:0009605                                      response ...        ...
GO:0051716 GO:0051716                                      cellular ...        ...
                  BgRatio       pvalue     p.adjust       qvalue
GO:0006955 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0002376 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0006952 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0050896 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0009605 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
GO:0051716 1347/16204 0.0000...  NA 0.0000...  NA 0.0000...  NA
```

5.4 Plot GO Results

```
dotplot(ego_bp, showCategory = 15)
```

6 Analysis 2: KEGG Enrichment

What is This? KEGG (Kyoto Encyclopedia of Genes and Genomes) is a "Christmas tree(pathway)" you can see represented in your significantly changed genes. It uses the KEGG pathway database which contains well-studied biological pathways.

KEGG Input Parameters Analysis

6.1 Convert SYMBOL to ENTREZ

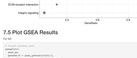
```
# Convert symbols to ENTREZ IDs
sig_genes <- ow_de[ow_de$padj < 0.05]
entrez_genes <- bitr(sig_genes,
                     fromType = "SYMBOL",
                     toType = "ENTREZID",
                     OrgDb = org.Hs.eg.db)

head(entrez_genes)
```

```
'select()' returned 1:1 mapping between keys and columns
Warning: Mapping via multiple columns failed.
Warning: 1 keys of input not mapped to ENTREZID.
  SYMBOL ENTREZID
NA     NA     <NA>
```

6.2 Run KEGG Enrichment

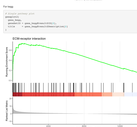
```
ekegg <- enrichKEGG(
  gene = entrez_genes$ENTREZID,
  organism = "hsa",
  pAdjustMethod = "BH",
  pvalueCutoff = 0.05)

head(ekegg@result)
```

```
                   ID                                        Description
hsa04060 hsa04060                 Cytokine-cytokine receptor interaction
hsa04640 hsa04640                                            Hematopoiet...
hsa04630 hsa04630                                            Jak-STAT sig...
hsa04514 hsa04514                                            Cell adhesion...
hsa04672 hsa04672                                            Intestinal i...
hsa04657 hsa04657                                            IL-17 signal...
hsa04668 hsa04668                                            TNF signaling...
hsa04659 hsa04659                                            Th17 cell di...
hsa04062 hsa04062                                            Chemokine sig...
hsa04061 hsa04061                                            Cytokine-cyto...
                 GeneRatio  BgRatio       pvalue     p.adjust       qvalue
hsa04060 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04640 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04630 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04514 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04672 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04657 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04668 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04659 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04062 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
hsa04061 .../...        ...  ...... 0.0000...  NA 0.0000...  NA 0.0000...  NA
                geneID Count
hsa04060 ...:...:...:
```



7.5 Plot GSEA Results

```
dotplot(
  gsea_go,
  showCategory = gsea_go$Description[1],
  title = gsea_go$Description[1]
)
```



For Kegg

```
dotplot(
  gsea_kegg,
  showCategory = gsea_kegg$Description[1],
  title = gsea_kegg$Description[1]
)
```

7.6 Extract Leading-Edge Genes

What is this? Genes driving the enrichment signal. Top genes that drive the enrichment signal. These are the most important genes in each pathway that contribute to the enrichment score.

```
# Extract leading-edge genes for GO
leading_go <- head(gsea_go@result, 5)
leading_go$c5.go.bp.leading_edge

head(leading_go, 2000)
```

```
##                   ID                                   Description
## GO:0006396 GO:0006396                                     RNA 3'-end processing
## GO:0006397 GO:0006397                                       mRNA processing
## GO:0006399 GO:0006399                                         tRNA metabolism
## GO:0006400 GO:0006400                 post-transcriptional modification: 2-thiouridine biosynthesis
## GO:0006401 GO:0006401                                       RNA catabolism
## GO:0006402 GO:0006402                                       mRNA catabolism
## GO:0006412 GO:0006412                                          translation
## GO:0006413 GO:0006413                                   translational initiation
## GO:0006414 GO:0006414                                   translational elongation
## GO:0006415 GO:0006415                                   translational termination
## GO:0006416 GO:0006416                                   translational fidelity
## GO:0006417 GO:0006417                                   regulation of translation
## GO:0006418 GO:0006418                                   tRNA aminoacylation for protein translation
## GO:0006419 GO:0006419                                   alanyl-tRNA aminoacylation
## GO:0006420 GO:0006420                                   arginyl-tRNA aminoacylation
## GO:0006421 GO:0006421                                   asparaginyl-tRNA aminoacylation
## GO:0006422 GO:0006422                                   aspartyl-tRNA aminoacylation
## GO:0006423 GO:0006423                                   cysteinyl-tRNA aminoacylation
## GO:0006424 GO:0006424                                   glutamyl-tRNA aminoacylation
## GO:0006425 GO:0006425                                   glutaminyl-tRNA aminoacylation
## GO:0006426 GO:0006426                                   glycyl-tRNA aminoacylation
## GO:0006427 GO:0006427                                   histidyl-tRNA aminoacylation
## GO:0006428 GO:0006428                                   isoleucyl-tRNA aminoacylation
## GO:0006429 GO:0006429                                   leucyl-tRNA aminoacylation
## GO:0006430 GO:0006430                                   lysyl-tRNA aminoacylation
## GO:0006431 GO:0006431                                   methionyl-tRNA aminoacylation
## GO:0006432 GO:0006432                                   phenylalanyl-tRNA aminoacylation
## GO:0006433 GO:0006433                                   prolyl-tRNA aminoacylation
## GO:0006434 GO:0006434                                   seryl-tRNA aminoacylation
## GO:0006435 GO:0006435                                   threonyl-tRNA aminoacylation
## GO:0006436 GO:0006436                                   tryptophanyl-tRNA aminoacylation
## GO:0006437 GO:0006437                                   tyrosyl-tRNA aminoacylation
## GO:0006438 GO:0006438                                   valyl-tRNA aminoacylation
## GO:0006439 GO:0006439                                   NADH metabolic process
## GO:0006440 GO:0006440                                   NADH biosynthesis
## GO:0006441 GO:0006441                                   NADH catabolism
## GO:0006442 GO:0006442                                   NADH oxidation
## GO:0006443 GO:0006443                                   NADH dehydrogenase (ubiquinone) activity
## GO:0006444 GO:0006444                                   NADH dehydrogenase (quinone) activity
## GO:0006445 GO:0006445                                   regulation of translation, ribosomal protein synthesis
## GO:0006446 GO:0006446                                   regulation of translation, initiation factor activity
## GO:0006447 GO:0006447                                   regulation of translation, peptide chain elongation factor activity
## GO:0006448 GO:0006448                                   regulation of translation, ribosomal protein synthesis
## GO:0006449 GO:0006449                                   regulation of translation, termination factor activity
## GO:0006450 GO:0006450                                   regulation of translation, tRNA binding
## GO:0006451 GO:0006451                                   regulation of translation, 5'-UTR binding
## GO:0006452 GO:0006452                                   regulation of translation, 3'-UTR binding
## GO:0006453 GO:0006453                                   regulation of translation, rRNA binding
## GO:0006454 GO:0006454                                   regulation of translation, tRNA binding
## GO:0006455 GO:0006455                                   regulation of translation, 5'-UTR binding
## GO:0006456 GO:0006456                                   regulation of translation, 3'-UTR binding
## GO:0006457 GO:0006457                                   regulation of translation, rRNA binding
## GO:0006458 GO:0006458                                   regulation of translation, tRNA binding
## GO:0006459 GO:0006459                                   regulation of translation, 5'-UTR binding
## GO:0006460 GO:0006460                                   regulation of translation, 3'-UTR binding
## GO:0006461 GO:0006461                                   regulation of translation, rRNA binding
## GO:0006462 GO:0006462                                   regulation of translation, tRNA binding
## GO:0006463 GO:0006463                                   regulation of translation, 5'-UTR binding
## GO:0006464 GO:0006464                                   regulation of translation, 3'-UTR binding
## GO:0006465 GO:0006465                                   regulation of translation, rRNA binding
## GO:0006466 GO:0006466                                   regulation of translation, tRNA binding
## GO:0006467 GO:0006467                                   regulation of translation, 5'-UTR binding
## GO:0006468 GO:0006468                                   regulation of translation, 3'-UTR binding
## GO:0006469 GO:0006469                                   regulation of translation, rRNA binding
## GO:0006470 GO:0006470                                   regulation of translation, tRNA binding
## GO:0006471 GO:0006471                                   regulation of translation, 5'-UTR binding
## GO:0006472 GO:0006472                                   regulation of translation, 3'-UTR binding
## GO:0006473 GO:0006473                                   regulation of translation, rRNA binding
## GO:0006474 GO:0006474                                   regulation of translation, tRNA binding
## GO:0006475 GO:0006475                                   regulation of translation, 5'-UTR binding
## GO:0006476 GO:0006476                                   regulation of translation, 3'-UTR binding
## GO:0006477 GO:0006477                                   regulation of translation, rRNA binding
## GO:0006478 GO:0006478                                   regulation of translation, tRNA binding
## GO:0006479 GO:0006479                                   regulation of translation, 5'-UTR binding
## GO:0006480 GO:0006480                                   regulation of translation, 3'-UTR binding
## GO:0006481 GO:0006481                                   regulation of translation, rRNA binding
## GO:0006482 GO:0006482                                   regulation of translation, tRNA binding
## GO:0006483 GO:0006483                                   regulation of translation, 5'-UTR binding
## GO:0006484 GO:0006484                                   regulation of translation, 3'-UTR binding
## GO:0006485 GO:0006485                                   regulation of translation, rRNA binding
## GO:0006486 GO:0006486                                   regulation of translation, tRNA binding
## GO:0006487 GO:0006487                                   regulation of translation, 5'-UTR binding
## GO:0006488 GO:0006488                                   regulation of translation, 3'-UTR binding
## GO:0006489 GO:0006489                                   regulation of translation, rRNA binding
## GO:0006490 GO:0006490                                   regulation of translation, tRNA binding
## GO:0006491 GO:0006491                                   regulation of translation, 5'-UTR binding
## GO:0006492 GO:0006492                                   regulation of translation, 3'-UTR binding
## GO:0006493 GO:0006493                                   regulation of translation, rRNA binding
## GO:0006494 GO:0006494                                   regulation of translation, tRNA binding
## GO:0006495 GO:0006495                                   regulation of translation, 5'-UTR binding
## GO:0006496 GO:0006496                                   regulation of translation, 3'-UTR binding
## GO:0006497 GO:0006497                                   regulation of translation, rRNA binding
## GO:0006498 GO:0006498                                   regulation of translation, tRNA binding
## GO:0006499 GO:0006499                                   regulation of translation, 5'-UTR binding
## GO:0006500 GO:0006500                                   regulation of translation, 3'-UTR binding
## GO:0006501 GO:0006501                                   regulation of translation, rRNA binding
## GO:0006502 GO:0006502                                   regulation of translation, tRNA binding
## GO:0006503 GO:0006503                                   regulation of translation, 5'-UTR binding
## GO:0006504 GO:0006504                                   regulation of translation, 3'-UTR binding
## GO:0006505 GO:0006505                                   regulation of translation, rRNA binding
## GO:0006506 GO:0006506                                   regulation of translation, tRNA binding
## GO:0006507 GO:0006507                                   regulation of translation, 5'-UTR binding
## GO:0006508 GO:0006508                                   regulation of translation, 3'-UTR binding
## GO:0006509 GO:0006509                                   regulation of translation, rRNA binding
## GO:0006510 GO:0006510                                   regulation of translation, tRNA binding
## GO:0006511 GO:0006511                                   regulation of translation, 5'-UTR binding
## GO:0006512 GO:0006512                                   regulation of translation, 3'-UTR binding
## GO:0006513 GO:0006513                                   regulation of translation, rRNA binding
## GO:0006514 GO:0006514                                   regulation of translation, tRNA binding
## GO:0006515 GO:0006515                                   regulation of translation, 5'-UTR binding
## GO:0006516 GO:0006516                                   regulation of translation, 3'-UTR binding
## GO:0006517 GO:0006517                                   regulation of translation, rRNA binding
## GO:0006518 GO:0006518                                   regulation of translation, tRNA binding
## GO:0006519 GO:0006519                                   regulation of translation, 5'-UTR binding
## GO:0006520 GO:0006520                                   regulation of translation, 3'-UTR binding
## GO:0006521 GO:0006521                                   regulation of translation, rRNA binding
## GO:0006522 GO:0006522                                   regulation of translation, tRNA binding
## GO:0006523 GO:0006523                                   regulation of translation, 5'-UTR binding
## GO:0006524 GO:0006524                                   regulation of translation, 3'-UTR binding
## GO:0006525 GO:0006525                                   regulation of translation, rRNA binding
## GO:0006526 GO:0006526                                   regulation of translation, tRNA binding
## GO:0006527 GO:0006527                                   regulation of translation, 5'-UTR binding
## GO:0006528 GO:0006528                                   regulation of translation, 3'-UTR binding
## GO:0006529 GO:0006529                                   regulation of translation, rRNA binding
## GO:0006530 GO:0006530                                   regulation of translation, tRNA binding
## GO:0006531 GO:0006531                                   regulation of translation, 5'-UTR binding
## GO:0006532 GO:0006532                                   regulation of translation, 3'-UTR binding
## GO:0006533 GO:0006533                                   regulation of translation, rRNA binding
## GO:0006534 GO:0006534                                   regulation of translation, tRNA binding
## GO:0006535 GO:0006535                                   regulation of translation, 5'-UTR binding
## GO:0006536 GO:0006536                                   regulation of translation, 3'-UTR binding
## GO:0006537 GO:0006537                                   regulation of translation, rRNA binding
## GO:0006538 GO:0006538                                   regulation of translation, tRNA binding
## GO:0006539 GO:0006539                                   regulation of translation, 5'-UTR binding
## GO:0006540 GO:0006540                                   regulation of translation, 3'-UTR binding
## GO:0006541 GO:0006541                                   regulation of translation, rRNA binding
## GO:0006542 GO:0006542                                   regulation of translation, tRNA binding
## GO:0006543 GO:0006543                                   regulation of translation, 5'-UTR binding
## GO:0006544 GO:0006544                                   regulation of translation, 3'-UTR binding
## GO:0006545 GO:0006545                                   regulation of translation, rRNA binding
## GO:0006546 GO:0006546                                   regulation of translation, tRNA binding
## GO:0006547 GO:0006547                                   regulation of translation, 5'-UTR binding
## GO:0006548 GO:0006548                                   regulation of translation, 3'-UTR binding
## GO:0006549 GO:0006549                                   regulation of translation, rRNA binding
## GO:0006550 GO:0006550                                   regulation of translation, tRNA binding
## GO:0006551 GO:0006551                                   regulation of translation, 5'-UTR binding
## GO:0006552 GO:0006552                                   regulation of translation, 3'-UTR binding
## GO:0006553 GO:0006553                                   regulation of translation, rRNA binding
## GO:0006554 GO:0006554                                   regulation of translation, tRNA binding
## GO:0006555 GO:0006555                                   regulation of translation, 5'-UTR binding
## GO:0006556 GO:0006556                                   regulation of translation, 3'-UTR binding
## GO:0006557 GO:0006557                                   regulation of translation, rRNA binding
## GO:0006558 GO:0006558                                   regulation of translation, tRNA binding
## GO:0006559 GO:0006559                                   regulation of translation, 5'-UTR binding
## GO:0006560 GO:0006560                                   regulation of translation, 3'-UTR binding
## GO:0006561 GO:0006561                                   regulation of translation, rRNA binding
## GO:0006562 GO:0006562                                   regulation of translation, tRNA binding
## GO:0006563 GO:0006563                                   regulation of translation, 5'-UTR binding
## GO:0006564 GO:0006564                                   regulation of translation, 3'-UTR binding
## GO:0006565 GO:0006565                                   regulation of translation, rRNA binding
## GO:0006566 GO:0006566                                   regulation of translation, tRNA binding
## GO:0006567 GO:0006567                                   regulation of translation, 5'-UTR binding
## GO:0006568 GO:0006568                                   regulation of translation, 3'-UTR binding
## GO:0006569 GO:0006569                                   regulation of translation, rRNA binding
## GO:0006570 GO:0006570                                   regulation of translation, tRNA binding
## GO:0006571 GO:0006571                                   regulation of translation, 5'-UTR binding
## GO:0006572 GO:0006572                                   regulation of translation, 3'-UTR binding
## GO:0006573 GO:0006573                                   regulation of translation, rRNA binding
## GO:0006574 GO:0006574                                   regulation of translation, tRNA binding
## GO:0006575 GO:0006575                                   regulation of translation, 5'-UTR binding
## GO:0006576 GO:0006576                                   regulation of translation, 3'-UTR binding
## GO:0006577 GO:0006577                                   regulation of translation, rRNA binding
## GO:0006578 GO:0006578                                   regulation of translation, tRNA binding
## GO:0006579 GO:0006579                                   regulation of translation, 5'-UTR binding
## GO:0006580 GO:0006580                                   regulation of translation, 3'-UTR binding
## GO:0006581 GO:0006581                                   regulation of translation, rRNA binding
## GO:0006582 GO:0006582                                   regulation of translation, tRNA binding
## GO:0006583 GO:0006583                                   regulation of translation, 5'-UTR binding
## GO:0006584 GO:0006584                                   regulation of translation, 3'-UTR binding
## GO:0006585 GO:0006585                                   regulation of translation, rRNA binding
## GO:0006586 GO:0006586                                   regulation of translation, tRNA binding
## GO:0006587 GO:0006587                                   regulation of translation, 5'-UTR binding
## GO:0006588 GO:0006588                                   regulation of translation, 3'-UTR binding
## GO:0006589 GO:0006589                                   regulation of translation, rRNA binding
## GO:0006590 GO:0006590                                   regulation of translation, tRNA binding
## GO:0006591 GO:0006591                                   regulation of translation, 5'-UTR binding
## GO:0006592 GO:0006592                                   regulation of translation, 3'-UTR binding
## GO:0006593 GO:0006593                                   regulation of translation, rRNA binding
## GO:0006594 GO:0006594                                   regulation of translation, tRNA binding
## GO:0006595 GO:0006595                                   regulation of translation, 5'-UTR binding
## GO:0006596 GO:0006596                                   regulation of translation, 3'-UTR binding
## GO:0006597 GO:0006597                                   regulation of translation, rRNA binding
## GO:0006598 GO:0006598                                   regulation of translation, tRNA binding
## GO:0006599 GO:0006599                                   regulation of translation, 5'-UTR binding
## GO:0006600 GO:0006600                                   regulation of translation, 3'-UTR binding
## GO:0006601 GO:0006601                                   regulation of translation, rRNA binding
## GO:0006602 GO:0006602                                   regulation of translation, tRNA binding
## GO:0006603 GO:0006603                                   regulation of translation, 5'-UTR binding
## GO:0006604 GO:0006604                                   regulation of translation, 3'-UTR binding
## GO:0006605 GO:0006605                                   regulation of translation, rRNA binding
## GO:0006606 GO:0006606                                   regulation of translation, tRNA binding
## GO:0006607 GO:0006607                                   regulation of translation, 5'-UTR binding
## GO:0006608 GO:0006608                                   regulation of translation, 3'-UTR binding
## GO:0006609 GO:0006609                                   regulation of translation, rRNA binding
## GO:0006610 GO:0006610                                   regulation of translation, tRNA binding
## GO:0006611 GO:0006611                                   regulation of translation, 5'-UTR binding
## GO:0006612 GO:0006612                                   regulation of translation, 3'-UTR binding
## GO:0006613 GO:0006613                                   regulation of translation, rRNA binding
## GO:0006614 GO:0006614                                   regulation of translation, tRNA binding
## GO:0006615 GO:0006615                                   regulation of translation, 5'-UTR binding
## GO:0006616 GO:0006616                                   regulation of translation, 3'-UTR binding
## GO:0006617 GO:0006617                                   regulation of translation, rRNA binding
## GO:0006618 GO:0006618                                   regulation of translation, tRNA binding
## GO:0006619 GO:0006619                                   regulation of translation, 5'-UTR binding
## GO:0006620 GO:0006620                                   regulation of translation, 3'-UTR binding
## GO:0006621 GO:0006621                                   regulation of translation, rRNA binding
## GO:0006622 GO:0006622                                   regulation of translation, tRNA binding
## GO:0006623 GO:0006623                                   regulation of translation, 5'-UTR binding
## GO:0006624 GO:0006624                                   regulation of translation, 3'-UTR binding
## GO:0006625 GO:0006625                                   regulation of translation, rRNA binding
## GO:0006626 GO:0006626                                   regulation of translation, tRNA binding
## GO:0006627 GO:0006627                                   regulation of translation, 5'-UTR binding
## GO:0006628 GO:0006628                                   regulation of translation, 3'-UTR binding
## GO:0006629 GO:0006629                                   regulation of translation, rRNA binding
## GO:0006630 GO:0006630                                   regulation of translation, tRNA binding
## GO:0006631 GO:0006631                                   regulation of translation, 5'-UTR binding
## GO:0006632 GO:0006632                                   regulation of translation, 3'-UTR binding
## GO:0006633 GO:0006633                                   regulation of translation, rRNA binding
## GO:0006634 GO:0006634                                   regulation of translation, tRNA binding
## GO:0006635 GO:0006635                                   regulation of translation, 5'-UTR binding
## GO:0006636 GO:0006636                                   regulation of translation, 3'-UTR binding
## GO:0006637 GO:0006637                                   regulation of translation, rRNA binding
## GO:0006638 GO:0006638                                   regulation of translation, tRNA binding
## GO:0006639 GO:0006639                                   regulation of translation, 5'-UTR binding
## GO:0006640 GO:0006640                                   regulation of translation, 3'-UTR binding
## GO:0006641 GO:0006641                                   regulation of translation, rRNA binding
## GO:0006642 GO:0006642                                   regulation of translation, tRNA binding
## GO:0006643 GO:0006643                                   regulation of translation, 5'-UTR binding
## GO:0006644 GO:0006644                                   regulation of translation, 3'-UTR binding
## GO:0006645 GO:0006645                                   regulation of translation, rRNA binding
## GO:0006646 GO:0006646                                   regulation of translation, tRNA binding
## GO:0006647 GO:0006647                                   regulation of translation, 5'-UTR binding
## GO:0006648 GO:0006648                                   regulation of translation, 3'-UTR binding
## GO:0006649 GO:0006649                                   regulation of translation, rRNA binding
## GO:0006650 GO:0006650                                   regulation of translation, tRNA binding
## GO:0006651 GO:0006651                                   regulation of translation, 5'-UTR binding
## GO:0006652 GO:0006652                                   regulation of translation, 3'-UTR binding
## GO:0006653 GO:0006653                                   regulation of translation, rRNA binding
## GO:0006654 GO:0006654                                   regulation of translation, tRNA binding
## GO:0006655 GO:0006655                                   regulation of translation, 5'-UTR binding
## GO:0006656 GO:0006656                                   regulation of translation, 3'-UTR binding
## GO:0006657 GO:0006657                                   regulation of translation, rRNA binding
## GO:0006658 GO:0006658                                   regulation of translation, tRNA binding
## GO:0006659 GO:0006659                                   regulation of translation, 5'-UTR binding
## GO:0006660 GO:0006660                                   regulation of translation, 3'-UTR binding
## GO:0006661 GO:0006661                                   regulation of translation, rRNA binding
## GO:0006662 GO:0006662                                   regulation of translation, tRNA binding
## GO:0006663 GO:0006663                                   regulation of translation, 5'-UTR binding
## GO:0006664 GO:0006664                                   regulation of translation, 3'-UTR binding
## GO:0006665 GO:0006665                                   regulation of translation, rRNA binding
## GO:0006666 GO:0006666                                   regulation of translation, tRNA binding
## GO:0006667 GO:0006667                                   regulation of translation, 5'-UTR binding
## GO:0006668 GO:0006668                                   regulation of translation, 3'-UTR binding
## GO:0006669 GO:0006669                                   regulation of translation, rRNA binding
## GO:0006670 GO:0006670                                   regulation of translation, tRNA binding
## GO:0006671 GO:0006671                                   regulation of translation, 5'-UTR binding
## GO:0006672 GO:0006672                                   regulation of translation, 3'-UTR binding
## GO:0006673 GO:0006673                                   regulation of translation, rRNA binding
## GO:0006674 GO:0006674                                   regulation of translation, tRNA binding
## GO:0006675 GO:0006675                                   regulation of translation, 5'-UTR binding
## GO:0006676 GO:0006676                                   regulation of translation, 3'-UTR binding
## GO:0006677 GO:0006677                                   regulation of translation, rRNA binding
## GO:0006678 GO:0006678                                   regulation of translation, tRNA binding
## GO:0006679 GO:0006679                                   regulation of translation, 5'-UTR binding
## GO:0006680 GO:0006680                                   regulation of translation, 3'-UTR binding
## GO:0006681 GO:0006681                                   regulation of translation, rRNA binding
## GO:0006682 GO:0006682                                   regulation of translation, tRNA binding
## GO:0006683 GO:0006683                                   regulation of translation, 5'-UTR binding
## GO:0006684 GO:0006684                                   regulation of translation, 3'-UTR binding
## GO:0006685 GO:0006685                                   regulation of translation, rRNA binding
## GO:0006686 GO:0006686                                   regulation of translation, tRNA binding
## GO:0006687 GO:0006687                                   regulation of translation, 5'-UTR binding
## GO:0006688 GO:0006688                                   regulation of translation, 3'-UTR binding
## GO:0006689 GO:0006689                                   regulation of translation, rRNA binding
## GO:0006690 GO:0006690                                   regulation of translation, tRNA binding
## GO:0006691 GO:0006691                                   regulation of translation, 5'-UTR binding
## GO:0006692 GO:0006692                                   regulation of translation, 3'-UTR binding
## GO:0006693 GO:0006693                                   regulation of translation, rRNA binding
## GO:0006694 GO:0006694                                   regulation of translation, tRNA binding
## GO:0006695 GO:0006695                                   regulation of translation, 5'-UTR binding
## GO:0006696 GO:0006696                                   regulation of translation, 3'-UTR binding
## GO:0006697 GO:0006697                                   regulation of translation, rRNA binding
## GO:0006698 GO:0006698                                   regulation of translation, tRNA binding
## GO:0006699 GO:0006699                                   regulation of translation, 5'-UTR binding
## GO:0006700 GO:0006700                                   regulation of translation, 3'-UTR binding
## GO:0006701 GO:0006701                                   regulation of translation, rRNA binding
## GO:0006702 GO:0006702                                   regulation of translation, tRNA binding
## GO:0006703 GO:0006703                                   regulation of translation, 5'-UTR binding
## GO:0006704 GO:0006704                                   regulation of translation, 3'-UTR binding
## GO:0006705 GO:0006705                                   regulation of translation, rRNA binding
## GO:0006706 GO:0006706                                   regulation of translation, tRNA binding
## GO:0006707 GO:0006707                                   regulation of translation, 5'-UTR binding
## GO:0006708 GO:0006708                                   regulation of translation, 3'-UTR binding
## GO:0006709 GO:0006709                                   regulation of translation, rRNA binding
## GO:0006710 GO:0006710                                   regulation of translation, tRNA binding
## GO:0006711 GO:0006711                                   regulation of translation, 5'-UTR binding
## GO:0006712 GO:0006712                                   regulation of translation, 3'-UTR binding
## GO:0006713 GO:0006713                                   regulation of translation, rRNA binding
## GO:0006714 GO:0006714                                   regulation of translation, tRNA binding
## GO:0006715 GO:0006715                                   regulation of translation, 5'-UTR binding
## GO:0006716 GO:0006716                                   regulation of translation, 3'-UTR binding
## GO:0006717 GO:0006717                                   regulation of translation, rRNA binding
## GO:0006718 GO:0006718                                   regulation of translation, tRNA binding
## GO:0006719 GO:0006719                                   regulation of translation, 5'-UTR binding
## GO:0006720 GO:0006720                                   regulation of translation, 3'-UTR binding
## GO:0006721 GO:0006721                                   regulation of translation, rRNA binding
## GO:0006722 GO:0006722                                   regulation of translation, tRNA binding
## GO:0006723 GO:0006723                                   regulation of translation, 5'-UTR binding
## GO:0006724 GO:0006724                                   regulation of translation, 3'-UTR binding
## GO:0006725 GO:0006725                                   regulation of translation, rRNA binding
## GO:0006726 GO:0006726                                   regulation of translation, tRNA binding
## GO:0006727 GO:0006727                                   regulation of translation, 5'-UTR binding
## GO:0006728 GO:0006728                                   regulation of translation, 3'-UTR binding
## GO:0006729 GO:0006729                                   regulation of translation, rRNA binding
## GO:0006730 GO:0006730                                   regulation of translation, tRNA binding
## GO:0006731 GO:0006731                                   regulation of translation, 5'-UTR binding
## GO:0006732 GO:0006732                                   regulation of translation, 3'-UTR binding
## GO:0006733 GO:0006733                                   regulation of translation, rRNA binding
## GO:0006734 GO:0006734                                   regulation of translation, tRNA binding
## GO:0006735 GO:0006735                                   regulation of translation, 5'-UTR binding
## GO:0006736 GO:0006736                                   regulation of translation, 3'-UTR binding
## GO:0006737 GO:0006737                                   regulation of translation, rRNA binding
## GO:0006738 GO:0006738                                   regulation of translation, tRNA binding
## GO:0006739 GO:0006739                                   regulation of translation, 5'-UTR binding
## GO:0006740 GO:0006740                                   regulation of translation, 3'-UTR binding
## GO:0006741 GO:0006741                                   regulation of translation, rRNA binding
## GO:0006742 GO:0006742                                   regulation of translation, tRNA binding
## GO:0006743 GO:0006743                                   regulation of translation, 5'-UTR binding
## GO:0006744 GO:0006744                                   regulation of translation, 3'-UTR binding
## GO:0006745 GO:0006745                                   regulation of translation, rRNA binding
## GO:0006746 GO:0006746                                   regulation of translation, tRNA binding
## GO:0006747 GO:0006747                                   regulation of translation, 5'-UTR binding
## GO:0006748 GO:0006748                                   regulation of translation, 3'-UTR binding
## GO:0006749 GO:0006749                                   regulation of translation, rRNA binding
## GO:0006750 GO:0006750                                   regulation of translation, tRNA binding
## GO:0006751 GO:0006751                                   regulation of translation, 5'-UTR binding
## GO:0006752 GO:0006752                                   regulation of translation, 3'-UTR binding
## GO:0006753 GO:0006753                                   regulation of translation, rRNA binding
## GO:0006754 GO:0006754                                   regulation of translation, tRNA binding
## GO:0006755 GO:0006755                                   regulation of translation, 5'-UTR binding
## GO:0006756 GO:0006756                                   regulation of translation, 3'-UTR binding
## GO:0006757 GO:0006757                                   regulation of translation, rRNA binding
## GO:0006758 GO:0006758                                   regulation of translation, tRNA binding
## GO:0006759 GO:0006759                                   regulation of translation, 5'-UTR binding
## GO:0006760 GO:0006760                                   regulation of translation, 3'-UTR binding
## GO:0006761 GO:0006761                                   regulation of translation, rRNA binding
## GO:0006762 GO:0006762                                   regulation of translation, tRNA binding
## GO:0006763 GO:0006763                                   regulation of translation, 5'-UTR binding
## GO:0006764 GO:0006764                                   regulation of translation, 3'-UTR binding
## GO:0006765 GO:0006765                                   regulation of translation, rRNA binding
## GO:0006766 GO:0006766                                   regulation of translation, tRNA binding
## GO:0006767 GO:0006767                                   regulation of translation, 5'-UTR binding
## GO:0006768 GO:0006768                                   regulation of translation, 3'-UTR binding
## GO:0006769 GO:0006769                                   regulation of translation, rRNA binding
## GO:0006770 GO:0006770                                   regulation of translation, tRNA binding
## GO:0006771 GO:0006771                                   regulation of translation, 5'-UTR binding
## GO:0006772 GO:0006772                                   regulation of translation, 3'-UTR binding
## GO:0006773 GO:0006773                                   regulation of translation, rRNA binding
## GO:0006774 GO:0006774                                   regulation of translation, tRNA binding
## GO:0006775 GO:0006775                                   regulation of translation, 5'-UTR binding
## GO:0006776 GO:0006776                                   regulation of translation, 3'-UTR binding
## GO:0006777 GO:0006777                                   regulation of translation, rRNA binding
## GO:0006778 GO:0006778                                   regulation of translation, tRNA binding
## GO:0006779 GO:0006779                                   regulation of translation, 5'-UTR binding
## GO:0006780 GO:0006780                                   regulation of translation, 3'-UTR binding
## GO:0006781 GO:0006781                                   regulation of translation, rRNA binding
## GO:0006782 GO:0006782                                   regulation of translation, tRNA binding
## GO:0006783 GO:0006783                                   regulation of translation, 5'-UTR binding
## GO:0006784 GO:0006784                                   regulation of translation, 3'-UTR binding
## GO:0006785 GO:0006785                                   regulation of translation, rRNA binding
## GO:0006786 GO:0006786                                   regulation of translation, tRNA binding
## GO:0006787 GO:0006787                                   regulation of translation, 5'-UTR binding
## GO:0006788 GO:0006788                                   regulation of translation, 3'-UTR binding
## GO:0006789 GO:0006789                                   regulation of translation, rRNA binding
## GO:0006790 GO:0006790                                   regulation of translation, tRNA binding
## GO:0006791 GO:0006791                                   regulation of translation, 5'-UTR binding
## GO:0006792 GO:0006792                                   regulation of translation, 3'-UTR binding
## GO:0006793 GO:0006793                                   regulation of translation, rRNA binding
## GO:0006794 GO:0006794                                   regulation of translation, tRNA binding
## GO:0006795 GO:0006795                                   regulation of translation, 5'-UTR binding
## GO:0006796 GO:0006796                                   regulation of translation, 3'-UTR binding
## GO:0006797 GO:0006797                                   regulation of translation, rRNA binding
## GO:0006798 GO:0006798                                   regulation of translation, tRNA binding
## GO:0006799 GO:0006799                                   regulation of translation, 5'-UTR binding
## GO:0006800 GO:0006800                                   regulation of translation, 3'-UTR binding
## GO:0006801 GO:0006801                                   regulation of translation, rRNA binding
## GO:0006802 GO:0006802                                   regulation of translation, tRNA binding
## GO:0006803 GO:0006803                                   regulation of translation, 5'-UTR binding
## GO:0006804 GO:0006804                                   regulation of translation, 3'-UTR binding
## GO:0006805 GO:0006805                                   regulation of translation, rRNA binding
## GO:0006806 GO:0006806                                   regulation of translation, tRNA binding
## GO:0006807 GO:0006807                                   regulation of translation, 5'-UTR binding
## GO:0006808 GO:0006808                                   regulation of translation, 3'-UTR binding
## GO:0006809 GO:0006809                                   regulation of translation, rRNA binding
## GO:0006810 GO:0006810                                   regulation of translation, tRNA binding
## GO:0006811 GO:0006811                                   regulation of translation, 5'-UTR binding
## GO:0006812 GO:0006812                                   regulation of translation, 3'-UTR binding
## GO:0006813 GO:0006813                                   regulation of translation, rRNA binding
## GO:0006814 GO:0006814                                   regulation of translation, tRNA binding
## GO:0006815 GO:0006815                                   regulation of translation, 5'-UTR binding
## GO:0006816 GO:0006816                                   regulation of translation, 3'-UTR binding
## GO:0006817 GO:0006817                                   regulation of translation, rRNA binding
## GO:0006818 GO:0006818                                   regulation of translation, tRNA binding
## GO:0006819 GO:0006819                                   regulation of translation, 5'-UTR binding
## GO:0006820 GO:0006820                                   regulation of translation, 3'-UTR binding
## GO:0006821 GO:0006821                                   regulation of translation, rRNA binding
## GO:0006822 GO:0006822                                   regulation of translation, tRNA binding
## GO:0006823 GO:0006823                                   regulation of translation, 5'-UTR binding
## GO:0006824 GO:0006824                                   regulation of translation, 3'-UTR binding
## GO:0006825 GO:0006825                                   regulation of translation, rRNA binding
## GO:0006826 GO:0006826                                   regulation of translation, tRNA binding
## GO:0006827 GO:0006827                                   regulation of translation, 5'-UTR binding
## GO:0006828 GO:0006828                                   regulation of translation, 3'-UTR binding
## GO:0006829 GO:0006829                                   regulation of translation, rRNA binding
## GO:0006830 GO:0006830                                   regulation of translation, tRNA binding
## GO:0006831 GO:0006831                                   regulation of translation, 5'-UTR binding
## GO:0006832 GO:0006832                                   regulation of translation, 3'-UTR binding
## GO:0006833 GO:0006833                                   regulation of translation, rRNA binding
## GO:0006834 GO:0006834                                   regulation of translation, tRNA binding
## GO:0006835 GO:0006835                                   regulation of translation, 5'-UTR binding
## GO:0006836 GO:0006836                                   regulation of translation, 3'-UTR binding
## GO:0006837 GO:0006837                                   regulation of translation, rRNA binding
## GO:0006838 GO:0006838                                   regulation of translation, tRNA binding
## GO:0006839 GO:0006839                                   regulation of translation, 5'-UTR binding
## GO:0006840 GO:0006840                                   regulation of translation, 3'-UTR binding
## GO:0006841 GO:0006841                                   regulation of translation, rRNA binding
## GO:0006842 GO:0006842                                   regulation of translation, tRNA binding
## GO:0006843 GO:0006843                                   regulation of translation, 5'-UTR binding
## GO:0006844 GO:0006844                                   regulation of translation, 3'-UTR binding
## GO:0006845 GO:0006845                                   regulation of translation, rRNA binding
## GO:0006846 GO:0006846                                   regulation of translation, tRNA binding
## GO:0006847 GO:0006847                                   regulation of translation, 5'-UTR binding
## GO:0006848 GO:0006848                                   regulation of translation, 3'-UTR binding
## GO:0006849 GO:0006849                                   regulation of translation, rRNA binding
## GO:0006850 GO:0006850                                   regulation of translation, tRNA binding
## GO:0006851 GO:0006851                                   regulation of translation, 5'-UTR binding
## GO:0006852 GO:0006852                                   regulation of translation, 3'-UTR binding
## GO:0006853 GO:0006853                                   regulation of translation, rRNA binding
## GO:0006854 GO:0006854                                   regulation of translation, tRNA binding
## GO:0006855 GO:0006855                                   regulation of translation, 5'-UTR binding
## GO:0006856 GO:0006856                                   regulation of translation, 3'-UTR binding
## GO:0006857 GO:0006857                                   regulation of translation, rRNA binding
## GO:0006858 GO:0006858                                   regulation of translation, tRNA binding
## GO:0006859 GO:0006859                                   regulation of translation, 5'-UTR binding
## GO:0006860 GO:0006860                                   regulation of translation, 3'-UTR binding
## GO:0006861 GO:0006861                                   regulation of translation, rRNA binding
## GO:0006862 GO:0006862                                   regulation of translation, tRNA binding
## GO:0006863 GO:0006863                                   regulation of translation, 5'-UTR binding
## GO:0006864 GO:0006864                                   regulation of translation, 3'-UTR binding
## GO:0006865 GO:0006865                                   regulation of translation, rRNA binding
## GO:0006866 GO:0006866                                   regulation of translation, tRNA binding
## GO:0006867 GO:0006867                                   regulation of translation, 5'-UTR binding
## GO:0006868 GO:0006868                                   regulation of translation, 3'-UTR binding
## GO:0006869 GO:0006869                                   regulation of translation, rRNA binding
## GO:0006870 GO:0006870                                   regulation of translation, tRNA binding
## GO:0006871 GO:0006871                                   regulation of translation, 5'-UTR binding
## GO:0006872 GO:0006872                                   regulation of translation, 3'-UTR binding
## GO:0006873 GO:0006873                                   regulation of translation, rRNA binding
## GO:0006874 GO:0006874                                   regulation of translation, tRNA binding
## GO:0006875 GO:0006875                                   regulation of translation, 5'-UTR binding
## GO:0006876 GO:0006876                                   regulation of translation, 3'-UTR binding
## GO:0006877 GO:0006877                                   regulation of translation, rRNA binding
## GO:0006878 GO:0006878                                   regulation of translation, tRNA binding
## GO:0006879 GO:0006879                                   regulation of translation, 5'-UTR binding
## GO:0006880 GO:0006880                                   regulation of translation, 3'-UTR binding
## GO:0006881 GO:0006
```

```
  }
}
```

```
## Saved: nes-distribution-gsea.png
```

10.13 12. Create Top Pathways Bar Chart

```
# Create bar chart of top pathways from each method
if(exists("comparison_df")) {
  # Get top 5 pathways from each method
  top_pathways <- comparison_df %>%
    group_by(Method) %>%
    top_n(5, wt = -Padj) %>%
    ungroup() %>%
    mutate(
      Pathway_short = ifelse(nchar(Pathway) > 50,
                             paste0(substr(Pathway, 1, 47), "..."),
                             Pathway)
    )

  p_top <- ggplot(top_pathways,
                  aes(x = reorder(Pathway_short, -log10Padj),
                      y = log10Padj,
                      fill = Method)) +
    geom_bar(stat = "identity", alpha = 0.8) +
    coord_flip() +
    facet_wrap(~ Method, scales = "free_y", ncol = 2) +
    scale_fill_brewer(palette = "Set2") +
    labs(
      title = "Top 5 Pathways by Method",
      x = "Pathway",
      y = "-log10(Adjusted P-value)"
    ) +
    theme_minimal() +
    theme(
      legend.position = "none",
      plot.title = element_text(hjust = 0.5, face = "bold"),
      axis.text.y = element_text(size = 8),
      strip.text = element_text(face = "bold")
    )

  ggsave(
    filename = file.path(blog_dir, "top-pathways-by-method.png"),
    plot = p_top,
    width = 14,
    height = 10,
    dpi = 300,
    bg = "white"
  )
  cat("Saved: top-pathways-by-method.png\n")
}
```

```
## Saved: top-pathways-by-method.png
```

10.14 Summary of Saved Images

```
##
## === Saved Blog Images ===
## Total images saved: 12
##
## - combined-comparison-plot.png
## - gene-count-comparison.png
## - go-enrichment-dotplot.png
## - gsea-go-dotplot.png
## - gsea-go-top-pathway.png
## - gsea-kegg-dotplot.png
## - gsea-kegg-top-pathway.png
## - kegg-enrichment-dotplot.png
## - method-summary-comparison.png
## - nes-distribution-gsea.png
## - pvalue-distribution-comparison.png
## - top-pathways-by-method.png
##
## All images saved to: blog_images
## You can now use these images in your blog post!
```